

VLADIMIR IVANOV

PROFESSIONAL STRUCTURAL ENGINEER | + 61 424 339 354 | VMIVANOV@YAHOO.COM

QUALIFICATIONS

- Master of Science in Civil Engineering, Massachusetts Institute of Technology, 2001
- Master of Structural Engineering, University of Architecture, Civil Engineering and Geodesy, 1997

LANGUAGES

- Bulgarian (native)
- English (fluent)
- Russian (fluent)
- French (basic)
- German (basic)

COMPUTER SKILLS

- Microsoft Windows & Office
- Microtran
- SpaceGass
- RAPT
- TEDDS
- Strand7
- Staad.Pro
- SAP2000, ETABS, CSI Bridge
- ROBOT
- SOLIDWORKS
- SAFE
- ANSYS
- ABAQUS
- Plaxis
- Bentley View
- AutoCAD
- ProgeCAD
- ZWCAD
- LIMCON

ADDITIONAL TRAINING

- Schlumberger Introduction to Oil Industry
- Concrete Structures to AS3600
- Steel Structures to AS4100
- Materials Storage & Handling
- Bentley Staad.Pro V8 Fundamentals
- Bentley Staad.Pro V8 Advanced Course
- ASI Specification & Performance of Bolting
- ASI Fabrication & Welding for Engineers
- Private study of engineering and construction methods, practices, technologies, approaches.
- Continuous development of structural engineering design tools, references and technical libraries

SUMMARY OF EXPERIENCE

Professional structural engineer with 25 years consulting experience

Sectors: Industrial; Oil, Gas & Energy; Maritime/Marine; Mining, Minerals and Metals; Materials Handling; Metallurgy; Heavy Machines; Water, Nuclear/Power; Commercial, Manufacturing, Residential, Health, Hospitality, Government, Research & Development, Materials Science.

Work locations: USA, EU, CIS, Australia

Project locations: EU, CIS, Australia, Africa, Asia, Pacific Islands

EXPERTISE & INTERESTS

- Engineering studies, Front End and Detailed Engineering & Design
- Site inspection, investigation, structural integrity assessment
- Asset management, infrastructure improvement, sustaining works
- Upgrading, strengthening, retrofit, repair, remedial works
- Finite element analysis, dynamic analysis, technical drafting, CAD/CAE
- Peer reviews and third-party inspections
- Construction supervision
- Steel, concrete, masonry, timber, aluminium, and composite structures
- Facilities, buildings, civil, underground, and machinery structures
- Tanks, gasholders, silos, bins, hoppers, etc.
- Boilers and pressure vessels, process piping, pipelines
- Supervision, training, and mentoring of junior team members
- Technical delivery of engineering and design projects
- Development of bids, proposals, cost and man-hour estimates
- Minerals & Metals, Materials Handling, Resources, Metallurgy
- Oil & Gas/Energy/Chemical
- Power, Water, Infrastructure
- Transportation, Communications
- Ports, Marine, Waterfront, Offshore & Subsea
- Residential, Commercial, Health, Hospitality, Government
- Industrial & Heavy Machinery Structures
- Research & Development

PROFESSIONAL AFFILIATION

- Chartered Structural Engineer, Engineers Bulgaria (2004)
- Association of Professional Engineers Australia (2013)
- Australian Steel Institute (2010)
- PIANC (2013)
- American Society of Civil Engineers (2002)
- European Federation of National Engineering Associations (FEANI)

ACHIEVEMENTS

- Designed many safe, fit-for purpose and cost-effective structures
- Provided value for money to clients and employers
- Pro-active result-oriented manner of work
- Efficient use of time, resources and materials
- Expert in Australian and international design codes and standards

SENIOR STRUCTURAL ENGINEER, EMTEK, 2023, PERTH WA

- **Mineral Resources Port Onslow Project** (detailed design: CV01 Frames, CV03 Belt Splicing Station, CV03 GTU & Drive Station, CV03 Head Chute, CV04 GTU & Drive Station, CV04 Head Chute, CV04 Head End Truss, CV05 Sampling Conveyor, Sampling Station. Checked: TS1 Chutes)
- **Rio Tinto Cape Lambert Port A Falling Object Protection Systems** (detailed design: Conveyor C198 Shedders)
- **Pilbara Minerals Pilgangoora Process Plant** (detailed design: Screen 14 Floor Modification, 251-MC-007 Shedder Roof)

CONSULTING STRUCTURAL ENGINEER, PRIVATE COMPANIES & CLIENTS, 2022, PERTH WA

- **Delorean Energy Victoria Project** (detailed structural design: Main Piperack and associated infrastructure)
- **Alcoa Fixed Plant Workshop TR5838** (detailed structural design: building steelwork and foundations, overhead crane-supporting structures)
- **BOC Gas Engineering Consultancy Support** (detailed structural design, lifting analysis, certification to DNV-ST-E273 and ISO 10855)
- **CAPS Compressor Lifting Frame Verification** (lifting analysis and certification to AS 4991)
- **Karara TSF Ramp Conveyor CV-101** (Head End inspection report, root cause analysis, defect remediation concept)
- **BHP Jimblebar & Warrawandu WTP MMF Access Platforms** (detailed design: 2 sets of modular platforms with stairways)
- **BHP POH Car Dumper CD4 Renewal** (detailed structural design: HPU Area, Substation & Transformer, Tie-In Platform, Cable Rack, Cell Guarding)

LEAD STRUCTURAL ENGINEER, LYCOPODIUM MINERALS, 2022, PERTH WA

- **Newmont Ghana Ahafo North Gold Project** (detailed design: Mill Building Steelwork, Footings and Suspended Slab)
- **Liontown Kathleen Valley Lithium Project** (detailed design: Micropile Footings, Primary Crushing General, Mill Building, WHIMS & TR Building)

SENIOR STRUCTURAL ENGINEER, JMD ENGINEERING, 2021 – 2022, PERTH WA

- **Alcoa Filter Press Plates Lifting Frame** (detailed design: custom-built WLL 5T lifting frame)
- **Alcoa Pinjarra B111 Crane Replacement** (detailed design: temporary runway for overhead crane replacement)
- **Bulong Gold Project** (detailed design: Milling Building, Mill Foundation and Feed Conveyor, Leaching and Adsorption Area)
- **CBH EGT Screen Replacement** (detailed design: Screen Supports and Access Platforms, Load Skating Frame, Lifting Trolley, beam strengthening)
- **Doral Minerals Floating Pumping Station** (detailed design: pontoons, access walkways, mooring system and foundations)
- **Iluka SR01 Plant Stack** (detailed design: Ø1750 and Ø750 stainless steel ducts, duct supports, stack and valve access platforms)
- **Iluka SR01 DSC Chain Conveyor** (detailed design: Kiln Inlet Building Level 1 Extension and Modification and Dump Bund for up to 1000°C conveyor)
- **Iluka Cataby MMU Grizzly** (detailed design: new removable grizzly for Mobile Mining Units)
- **Jaguar ROM Wall Remediation WP 1 and 2** (detailed design: ROM Wall remediation, new steel Apron Module)
- **KSS Trash Screen Upgrade** (detailed design: transfer tower modification and retrofit for new screen)
- **Norton Binduli Water Supply System** (detailed design of protective casings: Perth – Kalgoorlie Rail Line Crossing at Binduli)
- **Regis Resources Rosemont Primary Crusher Modifications** (detailed design: crusher support frame, transfer chute)
- **South32 FAC050 1st Stage Filtration Dilution WP2** (detail design: floor steelwork, removable filter hoods)
- **South32 FAC032 Caustic Heave** (detail design: new mezzanine; structural assessment: caustic heave of Filtration Building)
- **Telfer Pyrite Flotation Area Remediation** (detailed design: replacement and repair of heavily damaged concrete columns in Pyrite Flotation Area)
- **Talison Lab AC Platform** (detailed design: AC Platform steelwork, footings and associated works)
- **WAMMCO X-Ray Installation** (detailed design: safety-related steel structures).

SENIOR STRUCTURAL ENGINEER, ONYX PROJECTS, 2020 – 2021, PERTH WA

- **BHP Yandi Belt Feeder Maintenance Platform** (detailed design and lifting study: WLL 16T balanced removable modular platform)
- **BHP Eastern Ridge Conveyor RC03** (addition of walkways and platforms; concept design)
- **BHP Yandi OHP3 UPS Replacement** (structural assessment and strengthening; checking engineer)
- **BHP NiW Silo No.1 Dust Collector Removal** (lifting frame design)
- **BHP NiW NLN Magnetic Separator** (detailed design, lifting study: Mill Raft Foundation, Miscellaneous Concrete, Rejects Screen Chutes)
- **BHP Nelson Point Car Dumper 1 Cell Removal** (structural assessment: removal of CD1 Building roof, cyclonic conditions)
- **FMG Iron Bridge Magnetite Project** (detailed design: Port CHF Return Water Area, OPF Concentrate Handling and PD Pump Area)
- **Rio Tinto Brockman 4 Ammonium Nitrate Facility** (CAPEX PFS: Water Treatment, ANFO Facilities, AN Emulsion, LV Washdown, Magazines)
- **Rio Tinto Cape Lambert Port A Maintenance Workshop** (detailed design: MRC 55Tx22M Portal Crane runway)
- **Rio Tinto Hope Downs 1 NE2 Stage 3 Cutback FS** (detailed design: Pit Dewatering Pipeline Anchor Blocks)
- **Rio Tinto Nammuldi BWT Classification Building 1 & 2 Loading Bays** (detailed design: Loading Bay Platforms)
- **Rio Tinto Nammuldi BWT Conveyor CV2210 Belt Reeling Platform** (detailed design; checking engineer)
- **Rio Tinto Nammuldi BWT Cyclone Isolation Valve Upgrade** (detailed design; checking engineer)
- **Rio Tinto Nammuldi BWT Vibrating Feeder Platform** (detailed design: Feeder Platform Extension)

SENIOR STRUCTURAL ENGINEER, PRIMERO GROUP, 2019 – 2020, PERTH WA

- **Rio Tinto Koodaideri NPI** (detailed design: EMO Carpark; Data Centres; LME Workshop; HME Workshop, Warehouse, Office & Amenities; Fabrication Workshop; HME Tyre Change, Nitrogen Plant & Tool Store; Gas Bottle Storage; Bulk Lubrication Storage & Distribution Piperack; LME Washbay & Oily Water Treatment; HME Washbay & Oily Water Treatment; SP2 Water Services; Electrical Storage Shed; FPM SP3 Site General; FPM Workshop & Oily Water Treatment; SP3 Water Services; crane runway designs to AS 1418 for 5t, 15t, 2x35t, and 50t overhead travelling cranes).
- **Covalent Lithium ECI** (CAPEX design and MTO: Primary, Secondary and Tertiary Crushing; Ore Storage Shed and Reclaim Tunnel; Screening; Grinding & Classification; Separation & Flotation; Primary & Secondary Magnetic Separation; Tails Handling; Transfer Conveyors and Stations)
- **FMG Christmas Creek WHIMS Upgrade** (structural design: Screen Building; Filter Building; Support structures for Switchrooms SR501, 502 & 503)
- **FMG Christmas Creek Crushing Plant** (3rd party structural review of modular crushing plant: sizer, conveyors, transfer and take-up stations, bins)
- **Tronox Chandalla Process Plant** (remedial design: Dry Plant, Pugging Shed, Piperack, Tank TK-122, Rutile Plant)
- **Rio Tinto MESA J NPI** (construction support: Rescreening; Return Conveyor & TS; Piperacks; Thickener & Flocculation, Switchrooms; Area Lighting)
- **Rio Tinto MESA K NPI** (construction support: Ammonium Nitrate Storage)
- **Kalium Lakes Beyondie Lateral Inlet & Delivery Station** (detailed structural design: Pressure Regulation, Filtration, and Isolation areas)

SENIOR STRUCTURAL ENGINEER, WSP AUSTRALIA, 2018 – 2019, PERTH WA

- **Air Liquide Kwinana Substation ASU1** (design of supporting structure for existing site with limited access and space available)
- **Air Liquide Kwinana ASU3 LOX Tank** (concept design for CAPEX and fee estimate)
- **BHP NiW Leinster Bridge Crane Replacement** (checking engineer; structural assessment, concept remedial design)
- **BHP RTP Scope of Works** (concept design, modular structures for locomotive fleet upgrade)
- **CBH Horizontal Storage Modifications** (structural assessment and detailed design; multiple Elevator Towers and Transfer Galleries in WA)
- **CBH Mt Madden TEP** (structural concept design for CAPEX: Grain Elevator, Transfer Conveyor; Overhead Storage Conveyor)
- **CBH Esperance Shiploader** (concept design and checking: shiploader slew drive jacking frame)
- **CBH Esperance Renewable Energy PFS** (concept design and checking, biogas plant)
- **CBH Kwinana Grain Terminal Dust Suppression** (structural assessment and design: Oil Storage Area; Underground Conveyor Tunnel)
- **ENGIE Port of Albany Berth 2 Fuel Pellet Shiploader Platform** (detailed design: wharf modification; steel structures; concrete structures)
- **ILUKA Cataby IPT** (detailed design: Flocculant Area; Fire & Process Water; Sand Tails Return Tank; GDPL Barite Circuit; NCSM HMC Reveal)
- **KCGM CIL 2 Tank Platform** (structural assessment, inspection, surveying, detailed design)
- **RTIO Koodaideri NPI** (D&C tender design; HME Workshop, Warehouse & Office; FPM Workshop)
- **RTIO Cape Lambert Port A Magnet Upgrade** (conversion of transfer station into magnet-supporting structure)
- **RTIO West Pilbara** (structural assessment, detailed structural design: Primary Crusher Monorails 7A-7B; CV9 & CV10 Belt Reeling Frame)
- **RTIO Yandi Seizer Breaker** (checking, concept design: lightweight jacking frame)
- **RTIO Paraburdoo CV25 Belt Reeler** (checking engineer, structural assessment of existing belt reeler support frame)
- **RTIO West Angelas Primary Crusher Conveyor C001** (structural assessment of damaged existing truss)
- **RTIO Yandi Wet Screen Replacement** (detailed design of steel structures)
- **Rio Tinto DSL Twin Boom Stacker** (detailed design of concrete structures)

SENIOR STRUCTURAL ENGINEER, JACOBS AUSTRALIA, 2017 – 2018, PERTH WA

- **BHP South Flank Project** (detailed design: Culverts, Train Loadout TLO1 Modification; MC15 Shedder; Long-Span Boom, HF Chutes, Safety Barrier)
- **BHP Nelson Point Port Debottlenecking Project 2 IPS** (CAPEX concept design: P24, P201, P602, P513, P514, P770; TS25, TS201, TS513, TS601)
- **BHP Jimblebar Stretch Assist DPS** (Dump Hoppers & Feeders Building; Transfer Station; Conveyor CV183, Tramp Metal & Belt Turning Station)
- **BHP Newman Stretch Assist DPS** (structural engineering and design)
- **BHP Car Dumper CD2 Dust Collection – Modularisation** (Scrubber, Duct Bridge, Access, Northern Stair, and Southern Stair Modules)
- **RTIO Cape Lambert C214 Belt Reeling Frame** (detailed design);
- **RTIO Brockman 2 Fines Stacker 3001STK001** (tripper head chute upgrade)

SENIOR STRUCTURAL ENGINEER, THYSSENKRUPP INDUSTRIAL SOLUTIONS AUSTRALIA, 2017, PERTH WA

- **BHP Mt Arthur Raw Coal Reclaimer** (scraper conveyor beam replacement; detailed steelwork and fatigue design)
- **BHP Hay Pt Coal Terminal Stacker/Reclaimer SR5** (stability assessment)
- **FMG Anderson Point Port MRMP RC702, RC703, SL701 & SL703** (structural inspections, defects identification, risk assessment)
- **FMG Anderson Point Port Shiploaders SL701, SL702 & SL703 Crane Systems** (detailed steelwork design)
- **BHP Nelson Point Port Reclaimer RC01** (stability assessment)
- **BHP Nelson Point Port Reclaimer RC02** (conveyor structure strength assessment; balancing and stability analysis)
- **BHP Jimblebar Lump Stacker ST02** (transfer chute modifications, steelwork assessment)
- **BHP MAC Stacker ST03** (balancing and stability assessment)
- **BHP Finucane Island Stackers STK09 & STK10** (balancing and stability assessments)
- **BHP Nelson Point Port Reclaimer BWR6 BWCO** (Bucketwheel changeout; review and certification of BHP methodology; site engineering support)
- **BHP Nelson Point Port Reclaimer BWR7 Buckets** (new frame; structural design)
- **BHP Nelson Point Port Reclaimer BWR8 BWCO** (Bucketwheel changeout, review and certification of BHP methodology, site inspection)
- **TBPS Stacker Reclaimer SR001 SBCO, Malaysia** (slew bearing changeout; detailed steelwork design)

CONSULTING STRUCTURAL ENGINEER, PRIVATE CLIENTS, 2015 – 2016, PERTH WA

- **Golden Square 32 St George's Terrace** (structural/site engineer: design, fabrication, installation and testing of curtain walls and facades)
- **Building Extensions and Modifications, Perth** (detailed design: masonry, steel, timber, tensile, and concrete structures)
- **CBH Gairdner Elevator** (3D finite element analysis, steelwork and bored pier foundation design of new platform structure)
- **Information Exchange in Commercial and Infrastructure Construction** (consulting major construction products manufacturer)
- **Multi-Storey Building, Bookara WA** (concrete, steel and timber structure; detailed structural design)
- **LWHPC R&D Project, Henderson WA** (research, development and design of new lightweight high-performance concretes and structures)
- **Phosphates Bunker Roof & Conveyor Support, Christmas Island** (3D finite element analysis, checking of drawings and calculations)
- **Mincor Tipping Bund** (non-linear kinematic and structural analysis, safety-related structure)
- **Saudi Mining Company** (consulting, review of structural drawings)
- **AngloGold Ashanti Tertiary Crusher Replacement** (consulting, documents review, estimates)
- **Thirty Meter Telescope NFIRAOS, Mauna Kea, Hawaii** (refrigerated enclosure for high-precision optical system; aluminium-composite structure)
- **BHP Yandi II OHP Screening Platforms, Pilbara WA** (detailed structural design: modular screening floor deck)
- **BHP Wharf Berth D Ship Loader Girder, Port Hedland WA** (detailed structural design: jacking frame)
- **BHP Yandi OHP Crusher Maintenance Frame** (detailed structural design: new lifting method)
- **BHP Ore Car BRD Installation, Perth WA** (concept design: Ore Car Lifting, Transport, and Boggie Frames; Ore Car Body Stands)
- **BHP Jimblebar Primary Crushing, Pilbara WA** (detailed structural design: Rock Breaker Maintenance Platform)
- **BHP CV700 Trestle & Trusses Replacement, Port Hedland WA** (concept design and CAPEX: temporary supports, strengthening)
- **BHP Modular Eyewash Stations, Port Hedland WA** (concept design for CAPEX: welded skid modules with 10KL potable water tanks)
- **BHP Ore Handling Area C Primary Crushing, Pilbara WA** (concept design and CAPEX: ROM Bin Underchute beams)

- **PRO RCR Davit Arm, Perth WA** (detailed structural design: installing a new davit arm in an existing building)
- **Royal Perth Hospital Helicopter Landing Pad, Perth WA** (concept design, research & CAPEX MTO: aluminium helipad with fuel fire suppression)

LEAD STRUCTURAL ENGINEER, WSP | PARSONS BRINCKERHOFF, 2015, PERTH WA

- **BHP Kwinana Nickel Refinery** (site inspection and remedial design: Wet Metal Building, Sundeck, Reduction, Area, Claves Building)
- **DSL Lake Macleod Dry Salt Bin Vibrators** (dynamic analysis, vibration-induced fatigue analysis, structural details, drawing checking)
- **DSL Cape Cuvier Rill Bin Assessment** (new retaining wall concept)
- **RTIO Western Turner Syncline Drive Station Pulley Changeout** (adjustable belt pulling frame 300kN WLL; pulley lifting C-frame 250kN WLL)
- **RTIO Tom Price Primary Crusher PC2 and Vibrating Grizzlies VG1 & VG2** (structural integrity inspection, defect identification, RSW preparation)

SENIOR STRUCTURAL ENGINEER, IRON ROAD PTY LTD, 2014 – 2015, ADELAIDE SA

- **Tails Filter Building** (6 off. 53x13.5x34m high modules weighing 3200t each; steel structures and concrete foundation raft)
- **Rail Car Unloading Facility** (53x16.5x24m high module weighing 3600t; steel structure, dump hoppers and concrete foundation raft)
- **Concentrate Magnetic Separation Building** (3 off. 53x18.5x46m high modules each weighing 4400t; steel structure and concrete foundation raft)
- **Rougher Magnetic Separation Building** (3 off. 46.5x18.5x45m high modules each weighing 4400t; steel structure and concrete foundation raft)
- **Gravity Building** (3 off. 33.5x18.5x44m high modules each weighing 3300t; steel structure and concrete foundation raft)
- **Vertimill Building** (3 off. 40x14x25m high modules each weighing 3000t; steel structure and concrete foundation raft)
- **SAG Mill Dry End Structure** (3 off. concrete foundation and structure for 2,700t mill; site-built concrete footings and steel structure)
- **SAG Mill Wet End Module** (3 off. 43.5x23.35x10m high modules each weighing 1000t; steel structure, platework and concrete foundation raft)
- **Plant Piperacks** (steel and concrete structure design, and modularisation of +1,500m single and double deck piperacks, spans 21,24, 30 and 36m)
- **Tail Thickener Feed Pumps & Hoppers** (3 off. 53x18.5x35m high module weighing 2000t; steelwork, platework and foundation design)
- **Tail Thickener Pump Chamber** (3 off. 18x16x10m high module weighing 1400t; buried composite steel-concrete structure design)
- **Tail Thickener Access Tunnel** (3 off. 18x8x10m high module weighing 800t; buried composite steel-concrete structure design)
- **Stockyard Conveyors** (Stacker, Reclaimer, SAG Mill Feed; total length 4000m+; low level modules, 24m gantries, trestles, precast footings)
- **Concentrate Conveyors** (Transfer, Bypass Stockpile, Stacker, Reclaimer, Train Load Out; total length 7000m+; preliminary design, MTO).
- **Train Load Out Module** (18.5x15x23m high module weighing over 1000t each; steel structure, surge bin platework, concrete foundation raft)
- **Ball Mill Feed End Structure** (3 off. site-built structures consisting of mill foundation, pad footings and steel structure)
- **Services & Vacuum Plant Module** (3 off. 18.5x51x22m high modules weighing over 1000t each; steel structure, concrete foundation raft)
- **Modularisation Design Criteria and Specification** (review)

SENIOR STRUCTURAL ENGINEER, DEPARTMENT OF MINES & PETROLEUM, 2013, PERTH WA

Reviewed crane designs to ensure compliance with regulations and standards. Assisted in development of guidelines and registration procedures.

SENIOR STRUCTURAL ENGINEER, MSP ENGINEERING, 2012 – 2013, PERTH WA

- **Modular Scrubber Trommel, Australian Ilmenite Resources, NT** (modular skid-mounted steel structure; in-service, pulling and lifting)
- **Ta & WHIMS Project, Talison Lithium - Greenbushes OPS, WA** (Ta & WHIMS Building steel structure and foundation raft)
- **Wet Concentrator Plant, Image Resources, WA** (Cyclone Stacker, PCP/SCP Buildings, Attrition Scrubbers, Thickener Foundations)
- **MFU Wet Plant, MZI Resources, WA** (Modular skid-mounted structures: Dump Hopper, Feed Conveyor, Trommel Steelwork, Specifications)
- **Tormin Mineral Sands Project, MCL, South Africa** (Cyclone Tower, Secondary Concentrator Plant, Trash Screen, Zircon Bin and Storage)

SENIOR STRUCTURAL ENGINEER, PARSONS BRINCKERHOFF, 2012 – 2013, PERTH WA

Engineering design and checking of industrial and marine structures for **LANCO AUSTRALIA COAL EXPORT TERMINAL PROJECT, BUNBURY PORT, WA**

- **Site Visit** (meeting with Bunbury Port Authority representatives, inspection of existing basin and vicinity)
- **Participation in project management** (progress meetings and communication with team, client and other stakeholders)
- **Basis of Design for Marine Structures** (detailed design criteria for marine structures - steel, concrete and composite)
- **Specification for Marine Structures** (piles, anchors, steel, concrete, fenders, bollards, quick release hooks, cathodic protection, coatings)
- **Export Wharf** (detailed structural design of an anchored pile wall multi-purpose wharf with one ship loader for ships 51,000 to 118,000t dwt)
- **Mooring Dolphins** (detailed structural design of piled dolphin structures for 120t quick release hooks)
- **Anchored Sea Wall** (conceptual design of anchored retaining pile wall around existing lease boundary for future navigational access)
- **Ship Loader Feed Conveyor CV302 Gallery** (conceptual design or modularised steel structure and pile foundations)
- **Surge Bin BB-01** (conceptual design of 2800t surge bin steel structure and pile foundation)
- **Transfer Station TT-02** (conceptual design of steel structure and pile foundations)
- **Transfer Station TT-01** (mentoring and guiding the structural engineer developing the concept structural design)

SENIOR STRUCTURAL ENGINEER, AECOM AUSTRALIA PTY LTD, 2012, PERTH WA

Structural engineering and design of marine structures for the **Rio Tinto Simandou Port Project, Guinea, West Africa**

- **Pioneering and Material Offloading Wharf & Approach Jetties** (seismic analysis, connection design)
- **Morebaya River Export Wharf** (detailed design, steel structure 1200x29m, max 3 berths, 250,000t dwt bulkers, 2x12,500tph shiploaders)
- **Morebaya River Transfer Platform** (seismic verification, 96x65m steel grillage composite deck structure)
- **Morebaya River Approach Jetty** (seismic verification, 456m long jetty with 2 conveyors and a roadway)
- **Morebaya River Berthing & Mooring Dolphins** (structural design, steel structure - jacket option, for bulk carriers of up to 250,000t dwt)
- **Ile Kabak Export Wharf** (steel structure, 780x23m, max 4 berths, 2500,000t dwt bulk carriers, 2x12,500tph ship loaders)
- **Mentoring of structural engineers** (use of Eurocodes, seismic design of marine structures, soil-structure interaction)

SENIOR STRUCTURAL ENGINEER, LANGKILDE ENGINEERING SERVICES PTY LTD, 2011 – 2012, PERTH WA

- **Parker Point Iron Ore Car Dumper Modifications, WA** (detailed structural design, adjustable steel structure, mobile maintenance module)
- **Windimurra Vanadium Project, WA** (Pipe Supports & Bridge, Kiln Removal Platform, Vibrating Screen Support)
- **DSL Pond Zero Pump Station Upgrade, Dampier, WA** (Pre-Cast Retaining Walls, Heavy-Duty Culverts, Foundation Blocks, Spillway)
- **Talison Greenbushes OPS Chemical Plant Expansion Project, WA** (Stockpile, Milling & Classification, Flotation, and Filtration Areas).
- **Rio Tinto T&I NeXCon Testing Rig, Perth, WA** (Ground Modules, Head End Gallery, Tail Take-up, Soft-Load Station, Conveyor CV-01)
- **DSL Salt Stockpile Reclaim, Port Headland, WA** (Conveyors CV-501 & CV-502, Transfer Station)

SENIOR STRUCTURAL ENGINEER, AZTEC ANALYSIS/W&G ENGINEERS PTY LTD, 2010, ADELAIDE SA

- **OneSteel Sustaining Projects** (detailed design: Coke Ovens Quench Pond, Blast Furnace No.2 Crane Upgrade, Rolling Mill Bloom Crane Modification)
- **National Cranes Spreader Beam Analysis** (detailed design: fixed and adjustable spreaders and lifting beams)

SENIOR STRUCTURAL ENGINEER, WORLEYPARSONS SERVICES PTY LTD, 2010 – 2011, PERTH WA

- **Jack Hills Iron Ore Expansion Project BFS** (review of Structural Design Criteria)
- **Roy Hill Iron Ore Project, Port Hedland WA** (concept design & modularisation: Car Dumper, Shuttling Conveyor CV-201, Transfer Station TS-103, Screen Building; review of Concrete Specification)
- **MLTonkolili Iron Ore Project, Sierra Leone** (participated in peer review at 20% DES)
- **FMG Iron Ore Conveyor Underpass Protection, Pilbara WA** (checking of modified conveyor steelwork design)
- **FMG AMS Workshop Crane Assessment, Cloudbreak Iron Ore Mine, WA** (o/head crane report review, conceptual design)
- **FMG 155MPTA Port Expansion, WA** (Design Criteria – Structures, Transfer Station TS908, Standard Gantry Modules, Trestles & Foundations)
- **WorleyParsons Steelwork Standards Drawings** (notes, base plates, splices, simple and rigid bolted connections, welded connections).
- **Mentoring students** (load engineering, finite element modelling, steelwork and connection design).

SENIOR STRUCTURAL ENGINEER, LAING O'ROURKE CONSTRUCTION AUSTRALIA LTD, 2009, SYDNEY NSW

Structural engineering, design and checking of steel and concrete structures for the **BHP Iron Ore Rapid Growth Project 5 - Yandi, WA**

- **Belt Splicing Station** (detailed design, structural steelwork and foundations)
- **Crusher Discharge Chutes**, (detailed design, dynamically loaded platework and steelwork)
- **ROM Dump Hopper** (detailed design, reinforced concrete foundation raft; advised the structural engineer designing the steelwork)
- **Vehicle Shedders** (detailed design, plated steel frame structure and foundations)
- **Overland Conveyor Head End Gantry** (engineer checking, suspended steel structure, concrete supports and foundations)

SENIOR STRUCTURAL ENGINEER, BATEMAN ENGINEERING PTY LTD, 2007 – 2009, PERTH WA

- **Ban Phuc Nickel Project, Vietnam** (detailed design: Grinding & Classification Building, Ball Mill Area, Ore Storage & Reclaim Area)
- **Erdenet Heap Leach BFS** (concept design for CAPEX purposes)
- **Karara Iron Ore Project FEED, WA** (Concentrate Filtration, Tailings Filtration, Secondary Crushing, Stockpile Feed Conveyor and Reclaim Tunnel)
- **BHP LNO Storage Facility, WA** (checked foundations, retaining wall, trafficable slab; designed steel rack and retaining wall)
- **BHP LNO De-bottlenecking Phase I, WA** (detailed steelwork design, piperack extension)
- **Muliashi Copper Project, Zambia** (Primary & Secondary Crushing, Standard and Stockpile Feed Conveyor, Reclaim Chamber, Agglomeration, Screens, Administration, Crib Room, Ablutions, Workshop & Warehouse, Laboratory, CCR, Main Piperack).
- **Nippon N-CHLO Pilot Plant, WA** (Piperack, Grinding, Leaching, Electrowinning, Carbon Columns, Screens, Leach Tanks, Mixers, Filter Presses)
- **Rio Tinto Pilot Iron Ore Sorter Plant, WA** (CAPEX MTOs, Ore Sorter Building, Scope of Works, Concrete Specification, IFC design approval)
- **Teramin Tala Hamza Zink & Lead PFS, Algeria** (concept design for CAPEX purposes)
- **Rio Tinto Aluminium Yavun 2 Area 720 – Boiler 3 Stack** (structural design criteria, basic design)
- **Sukari Gold Project, Egypt** (Reclaim, Grinding, Tunnel, Gold Room, Main Piperack, Stockpile Bunker, CIL Tank, Flotation & Re grind Thickeners)

SENIOR STRUCTURAL ENGINEER, AUSENCO SERVICES PTY LTD, 2006 – 2007, PERTH WA

- **Sabodala Gold Project, Senegal** (developed civil/structural MTOs for CapEx estimate; detailed structural design of Stockpile Feed Conveyor, Reclaim Chamber, SAG and Ball Mill Foundations, CIL Tanks and Platforms, Grinding/Gravity Circuit Area Building)
- **Lumwana Copper Project, Zambia** (detailed design of steelwork and foundations, Desorption Building)
- **Jinfeng Gold Project, China** (detailed design, Retaining Wall & Pipeline Anchor Block)
- **Minara Nickel Project, WA** (Dump Hopper: site inspection, damage assessment, design of composite retaining wall for impact protection)

PRINCIPAL STRUCTURAL ENGINEER, CHIMREMONTSTROY ENGINEERING JSC, 2006, BULGARIA & AUSTRALIA

- **HRSE Multi-Storey Office & Commercial Building** (detailed design and construction supervision, concrete structure)
- **Industrial Export Bio Diesel & Vegetable Oil Plant** (detailed structural & fire safety design, assistance with commissioning)
- **Chelopech Mining 3x500t Cement Silos** (conceptual design, supervision of detailed design, checking, design authorisation)
- **Sofia France Auto Trade & Expo Centre** (civil/structural study and CapEx estimate for tendering purposes)
- **Cumerio Pirdop Copper Smelter Cooling Spray Chamber** (detailed steelwork design)
- **HRSE Multi-Storey Residential & Commercial Building** (detailed design, concrete structure)
- **Sidenor/Stomana Industry Scrap Yard Reconstruction** (basic engineering, reconstruction of a 10,000 m² steel scrap yard)
- **Mital Steel Kremikovtzi Boiler K100 Steelwork Rehabilitation** (site inspection, damage assessment, strengthening design for a steam boiler)
- **Dana Trading Centre Project** (detailed design, composite steel-concrete structure)
- **Prosal Cold-Formed Welded Steel Tubes Plant** (site inspection, engineering and design supervision, industrial steel structure)
- **Krumovgrad Gold Project** (client's representative, conducted peer review of civil/structural design by Ausenco Services Pty Ltd, Perth WA)
- **Chelopech Mining Plant Expansion** (client's representative, conducted peer review of civil/structural design by Minrpoc Pty Ltd, Perth, WA)
- **Chelopech Mining Tailings Dam Drainage Project** (detailed design: Dewatering Shaft)
- **Unior Minier Air Compressors** (structural damage assessment, in-situ strength testing, supervision of demolition and re-construction)
- **Monastery Hotel & Villas Leisure Complex** (multi-storey hotel, villas, service buildings, church, football field, swimming pools, wastewater treatment)

SENIOR STRUCTURAL ENGINEER, EQE BULGARIA / AMERICAN BUREAU OF SHIPPING, 2003 – 2006, BULGARIA

- **Pumping Station and Circulation Pipelines, Unior Minier Pirdop Copper** (detailed design of steelwork, concrete structures and foundations; specifications, MTOs; site inspections, designer's construction supervision)
- **Block Batteries and DC Units, Kozloduy Nuclear Power Plant** (conceptual design; steel, concrete and masonry structures)
- **MCR Structures Seismic Assessment, Units 3 and 4, Kozloduy Nuclear Power Plant** (site inspection, integrity assessment, detailed design)
- **Overhead Cranes and Runway Beams in Units 3 and 4, Kozloduy Nuclear Power Plant** (structural integrity assessment of runways for overhead cranes with 30, 125 and 250 t capacity, in Turbine and Reactor Units; detailed engineering design)
- **Gas Fire Extinguishing System in Units 5 and 6, Kozloduy Nuclear Power Plant** (Lead Structural Engineer; site inspections, detailed design)
- **Forced Venting System in Units 3 and 4, Kozloduy Nuclear Power Plant** (Lead Structural Engineer; site inspections, detailed design)
- **Wastewater Treatment Plant, Unior Minier Pirdop Copper** (detailed engineering design, steel structures)
- **Boiler Steel Structure Integrity Assessment, Thermal Power Plant Maritza Iztok 2** (site inspection, bid preparation, detailed design)
- **Dry Storage for Spent Nuclear Fuel, Kozloduy Nuclear Power Plant** (standards reconciliation, design review; 1300m² piled raft)
- **Mall of Sofia, Block 3** (peer review of detailed structural design of the largest retail centre in Bulgaria; recommended design improvements)
- **Seismic Assessment of Business Park Sofia Buildings 7 and 9, Lindner Bulgaria** (bid preparation, site inspection, peer review)
- **Strength Assessment of Safety-Related Steam Pipelines in Units 5 and 6, Kozloduy Nuclear Power Plant** (inspection, remedial design)
- **Demolition and Removal of Elevated Feed Conveyor, Thermal Power Plant Maritza Iztok 2** (site inspection, bid preparation)
- **Second Report of the Republic of Bulgaria on the Joint Convention on Safe Usage of Nuclear Energy, BNRA** (translation)
- **Creation and management of Electronic Technical Library** (comprehensive collection of engineering books, standards, codes, and design tools)

STRUCTURAL ENGINEER, VOEST ALPINE INDUSTRIEANLAGENBAU (VAI), 2004 – 2005, UKRAINE & AUSTRIA

- **Erection of Ladle Furnace and Continuous Casting Machine in AMK** (site management: equipment certification, installation, testing and commissioning; structural design; defect rectification; inspection of fabrication plants; supervision of design teams in Ukraine; site office record keeping)

STRUCTURAL ENGINEER, BETA CONST, 2002 – 2003, SOFIA, BULGARIA

Structural engineering and design for residential, commercial and health projects

- **Tokuda Hospital Sofia, Tokushukai Medical Corporation** (participated in detailed design, composite concrete structures for the 85,000m² state-of-the-art general hospital of the largest private healthcare provider in the world)
- **Vesta-Gas Modular Gas Station, Cafeteria & Auto Store** (detailed design of modular tubular 3D truss, steelwork & footings)
- **Asenov Residence** (detailed design: luxurious manor in the outskirts of Vitosha Mountain)
- **Meir Ovadia Residential Building** (detailed design: composite steel – timber structure)
- **Adonis Residential Building** (detailed design: reinforced concrete structure)

STRUCTURAL ENGINEER, BORA LTD, 2001-2002, SOFIA, BULGARIA

Structural engineering and design of steel and reinforced concrete structures for wastewater treatment plants (WWTP). Projects:

- **Modernization and Extension of WWTP Sliven**
- **Construction of WWTP Sinemoretz**

DRILLING & MEASUREMENTS ENGINEER IN TRAINING, SCHLUMBERGER OILFIELD SERVICES, OKLAHOMA CITY, 2001, USA

Trained in oilfield HSE, introduction to oilfield services, safe driving. Duties: maintenance, delivery, use of drilling and measurement equipment.

RESEARCH ASSISTANT, MASSACHUSETTS INSTITUTE OF TECHNOLOGY, 1999 – 2001, CAMBRIDGE, MA, USA

Advanced scientific studies in the areas of structural, geotechnical, civil and environmental engineering.

- **Mt Ambine Base Tunnel Project, France/Switzerland** (geological conditions assessment; preliminary permanent concrete liner design)

STRUCTURAL ENGINEER, BORA LTD, 1998-1999, SOFIA, BULGARIA

Structural engineering and design of steel and reinforced concrete structures for the **Modernisation and Rehabilitation of WWTP Sofia**

CONSULTING STRUCTURAL ENGINEER, VARIOUS PRIVATE CLIENTS, 1997 – 1998, SOFIA, BULGARIA

Consulting and design: civil/structural engineering, construction materials and technology, asset management.

- **Multi-Purpose Aircraft Hangar** (concept structural design of a 96 x 70 m aircraft hangar)
- **Nikolova Multi-Storey Residential Building** (detailed design in all disciplines)
- **Dulev Multi-Storey Residential Building** (detailed design in all disciplines)
- **Kolesov Multi-Storey Residential Building & Dental Clinic** (detailed design in all disciplines)
- **Zarkov Multi-Storey Residential Building** (detailed design in all disciplines)
- **Kostov Multi-Storey Residential Building** (concept design)
- **PIMB Pavilion** (detailed design in all disciplines)
- **Kokalyane Landslide Mitigation Project** (detailed design: cantilever retaining wall)
- **Ramstore Cafeteria** (structural assessment, certification)